Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

# ENVIRONMENTAL SCIENCE

## Paper 1 (Generated practice paper - Set 32)

Assessed: The physical environment, Energy resources, Pollution, Research methods

**Time allowed: 3 hours**

### Materials

For this paper you may use:
- a calculator.

### Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

### Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

**For Examiner's Use**

| Question | Mark |
|---|---|
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |
| 6 | |
| 7 | |
| 8 | |
| 9 | |
| 10 | |
| 11 | |
| TOTAL | |

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**Answer all questions in the spaces provided.**

**0 1** Table 1 shows measurements taken from a stream above and below the point where drainage water from an old metal-mine spoil heap enters it.

**Table 1**

| Site | pH | Dissolved zinc / mg $dm^{-3}$ | Dissolved oxygen / % saturation | Number of invertebrate taxa | Mayfly larvae present? |
|---|---|---|---|---|---|
| Upstream | 7.1 | 0.02 | 96 | 18 | Yes |
| 50 m downstream | 3.4 | 6.5 | 94 | 3 | No |
| 2 km downstream | 5.9 | 1.1 | 95 | 9 | No |

**0 1 . 1** Explain how the acidic drainage water is produced in the spoil heap.

**[3 marks]**

**0 1 . 2** Use Table 1 to explain why the number of invertebrate taxa falls so sharply 50 m downstream, even though dissolved oxygen is unchanged.

**[3 marks]**

**0 1 . 3** Suggest why the pH and zinc concentration have partly recovered 2 km downstream.

**[2 marks]**

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**0 1 . 4** Describe how acid mine drainage can be controlled.

**[2 marks]**

**10**

**0 2 . 1** Explain how hydrogen can be used to store energy from intermittent renewable resources, and describe **two** ways the stored energy can be released.

**[4 marks]**

**0 2 . 2** Compare **three** methods of storing hydrogen, giving a disadvantage of each.

**[3 marks]**

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**0 2 . 3** Explain how a fuel cell differs from a rechargeable battery.

**[2 marks]**

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**0 2 . 4** Explain what is meant by the 'hydrogen economy'.

**[1 mark]**

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**10**

**0 3 . 1** Which chemical form of mercury is most easily absorbed and biomagnified in food chains? Tick **one** box.

**[1 mark]**

A Elemental (liquid) mercury ☐

B Inorganic mercury compounds ☐

C Organic (methyl) mercury ☐

D Mercury vapour ☐

**0 3 . 2** Explain why anaerobic sediments increased the hazard of the inorganic mercury discharged at Minamata.

**[2 marks]**

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**0 3 . 3** Explain why heavy metal wastes are stored under alkaline conditions.

**[2 marks]**

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5

**0 4 . 1** Explain how climate change may affect the survival of species. Use named examples.

**[6 marks]**

**0 4 . 2** Explain why a species living in an interdependent community may be unable to move to a new area even if suitable conditions exist there.

**[2 marks]**

**0 4 . 3** Explain why polar regions warming faster than the equator affects the weather in the UK.

**[2 marks]**

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**10**

**0 5** Table 1 shows three investigations carried out by a class.

**Table 1**

| Investigation | Data collected |
|---|---|
| A | Mean height of bluebells in a coppiced and an uncoppiced wood (both normally distributed) |
| B | Number of limpets in each of five zones on a rocky shore, compared with the numbers expected if they were evenly spread |
| C | Soil moisture and the percentage cover of a moss, measured at 14 points along a transect |

**0 5 . 1** Name the most suitable statistical test for each investigation in Table 1.

**[3 marks]**

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**0 5 . 2** Explain what it means if a test gives a probability of $p = 0.02$.

**[2 marks]**

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**0 5 . 3** Explain why the calculated value from a test must be compared with a critical value, and what else is needed to find that critical value.

**[3 marks]**

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**0 5 . 4** Explain what the **standard deviation** of each set of bluebell heights tells the student, and how it is shown on a graph of the means.

**[3 marks]**

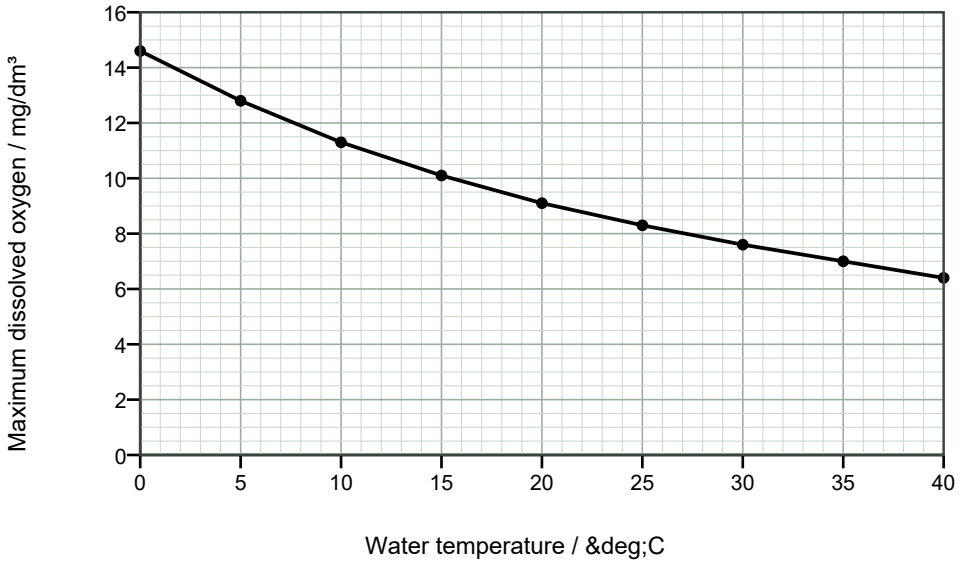
**0 5 . 5** The class found a significant correlation in investigation C. Explain **four** reasons why this does not show that soil moisture causes the moss to grow.

**[4 marks]**

**15**

**0 6** Figure 1 shows the maximum concentration of dissolved oxygen in fresh water at different temperatures. A power station discharges warm cooling water into a river. The lowest dissolved oxygen concentration that trout can tolerate is 6.5 mg $dm^{-3}$.

**Figure 1**




**0 6 . 1** Describe the relationship shown in Figure 1.

**[1 mark]**

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**0 6 . 2** Use Figure 1 to determine the highest water temperature at which trout could survive if the water were fully saturated with oxygen.

**[1 mark]**

**Answer** °C

**0 6 . 3** In summer the river upstream of the power station is at 20 °C and the effluent raises the temperature by 7 °C. Use Figure 1 to calculate the reduction in maximum dissolved oxygen concentration caused by the effluent.

**[2 marks]**

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**Answer** _____ mg/dm³

**0 6 . 4** Explain why the actual dissolved oxygen concentration downstream may be much lower than the values in Figure 1.

**[2 marks]**

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**0 6 . 5** Describe **three** effects of warmer water on aquatic organisms other than reduced dissolved oxygen.

**[3 marks]**

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**0 6 . 6** Describe how a cooling tower reduces the temperature of power station effluent water.

**[1 mark]**

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**0 7 . 1** Which of the following statements about stratospheric ozone is correct? Tick one box.

**[1 mark]**

A Ozone is destroyed only by CFCs ☐

B Ozone absorbs mainly infrared radiation ☐

C Ozone is formed when monatomic oxygen reacts with diatomic oxygen ☐

D Ozone concentrations are highest at ground level ☐

**0 7 . 2** Explain the difference between the roles of ozone in the stratosphere and in the troposphere.

**[2 marks]**

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**0 7 . 3** Explain why HCFCs cause less ozone depletion than CFCs.

**[2 marks]**

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**0 8 . 1** Give **three** advantages of incinerating domestic waste.

**[3 marks]**

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**2** .....

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**3** .....

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**0 8 . 2** Explain how each of the following features of a modern incinerator reduces pollution.

**[4 marks]**

Do not write outside the box

**High combustion temperature with rapid cooling of gases** .....

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**Water spray scrubber** .....

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**Removal of wet / compostable waste** .....

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**Removal of recyclable waste** .....

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**0 8 . 3** Explain why the ash from waste incinerators needs careful disposal.

**[2 marks]**

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**0 8 . 4** State **one** reason why incineration is expensive.

**[1 mark]**

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**10**

**0 9** Table 1 shows the effect of installing water meters on the mean daily water use of households in a UK town.

**Table 1**

| | **Unmetered households (fixed charge)** | **Metered households** |
|---|---|---|
| Mean water use / $dm^3$ per person per day | 150 | 125 |
| Percentage with low-flush toilets | 35 | 58 |
| Percentage using a garden hosepipe weekly in summer | 44 | 20 |

**0 9 . 1** Calculate the percentage reduction in water use per person in metered households.

**[1 mark]**

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**Answer** [ ] %

**0 9 . 2** Explain why metering reduces water use.

**[2 marks]**

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**0 9 . 3** Describe **three** ways in which domestic water use can be reduced other than metering.

**[3 marks]**

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**2** .....

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**3** .....

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**0 9 . 4** Explain why about 20% of the water intended for public supply in the UK never reaches consumers, and how this could be reduced.

**[2 marks]**

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**0 9 . 5** Explain how drip irrigation reduces the water needed to grow a crop compared with spray irrigation.

**[2 marks]**

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**1 0 . 1** Discuss how an understanding of the properties of pollutants can be used to predict their behaviour in the environment and to plan strategies to control them.

**[9 marks]**

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Extra space

**1 0 . 2** State what is meant by **bioaccumulation**.

**[1 mark]**

**10**

**Write an essay on one of the following topics.**

**1 1 . 1** Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably.

**[25 marks]**

**OR**

**1 1 . 2** Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it.

**[25 marks]**

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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**END OF QUESTIONS**